A proof searching procedure for intuitionistic propositional logic in Agda

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Motivation

- Problem: Mathematicians don't want to be wrong

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  Claim: All horses are the same color.[1]

  Proof (induction):

  Base case, one horse: Trivially true.

  Inductive step: Assume n horses are all the same color. Consider $n+1$ horses.

  First n horses are the same color.

  Last n horses are also the same color.

  Therefore all $n+1$ horses must be the same color.

[1] George Pólya, 1954. "Are any n numbers equal?"

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- Problem: Mathematicians are lazy
- Solution: Proof searching!

Outline

1. About Agda
2. Propositional formulas & valuations
3. Sequent calculus for IPC (LJ)
4. Formalizing Sequent calculus in Agda
5. Structural properties of Sequent calculus
6. Proof of soundness in Agda
7. Refinement of LJ – LJT and LJf
8. LJf is decidable – there is a search procedure for LJf
9. Proving termination in Agda
10. Translating LJf to LJ
11. Future work (WIP) LJ is decidable / LJf is complete

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Why Agda?

Propositional formulas & valuations

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1. Propositional variables
   P_1, P_2, P_3, $\cdots$

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$$A \vee B$$

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$$A \to B$$

$$A, B \text{ propositions}$$

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$$A \vee B$$

$$A \wedge B$$

$$A \to B$$

$$A, B \text{ propositions}$$

$$\neg A := A \longrightarrow \bot \qquad A \leftrightarrow B := A \to B \wedge B \to A$$

Valuations

$$a : \mathsf{Prop} \to \mathbb{B}_2 = \{0, 1\}$$

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$$
\begin{aligned}
[\![P_i]\!]^a &= a(i) \\
[\![\top]\!]^a &= 1 \\
[\![\bot]\!]^a &= 0 \\
[\![A \vee B]\!]^a &= [\![A]\!]^a \vee_{\mathbb{B}} [\![B]\!]^a \\
[\![A \wedge B]\!]^a &= [\![A]\!]^a \wedge_{\mathbb{B}} [\![B]\!]^a \\
[\![A \to B]\!]^a &= [\![A]\!]^a \to_{\mathbb{B}} [\![B]\!]^a
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\end{aligned}
$$

!! This is a classical semantics !!

Sequent calculus

Sequent calculus – Sequents

The anatomy of a sequent

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The anatomy of a sequent

$$\underbrace{A_1, A_2, \cdots, A_n}_{\text{Antecendents / Context}} \quad \vdash \quad \overset{\text{succedent}}{\tilde{C}} \qquad\qquad A_i, C \in \text{Prop}$$

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"Assuming $A_1, A_2, \cdots, A_n$ we can derive C"

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These are also sequents

$\Gamma \vdash C$

$\Gamma, A \vdash C$

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Note

- Order does not matter $X, Y, Z \vdash C \equiv Z, Y, X \vdash C$

Sequent calculus – Sequents

The anatomy of a sequent

$$\underbrace{A_1, A_2, \cdots, A_n}_{\text{Antecendents / Context}} \quad \vdash \quad \overset{\text{succedent}}{\tilde{C}} \qquad\qquad A_i, C \in \text{Prop}$$

"Assuming $A_1, A_2, \cdots, A_n$ we can derive C"

These are also sequents

$\Gamma \vdash C$

$\Gamma, A \vdash C$

Note

- Order does not matter $X, Y, Z \vdash C \equiv Z, Y, X \vdash C$
- Multiplicity matters $Z \vdash C \not\equiv Z, Z, Z \vdash C$

Sequent calculus – Inference rules

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Like natural deduction

$$\frac{\text{Premise}_1 \quad \text{Premise}_1 \quad \cdots \quad \text{Premise}_1}{\text{Conclusion}}\,\text{Label}$$

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$$A \wedge B \vdash B \wedge A$$

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$$A \wedge B \vdash B \wedge A$$

$$\frac{\Gamma \vdash A \quad \Gamma \vdash B}{\Gamma \vdash A \wedge B}\,\wedge_R$$

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$$\frac{\Gamma, A \wedge B, B \vdash C}{\Gamma, A \wedge B \vdash C}\,\wedge_{L_2}$$

$$\frac{\Gamma, A \wedge B, A \vdash C}{\Gamma, A \wedge B \vdash C}\,\wedge_{L_1}$$

$$\frac{\Gamma \vdash A \quad \Gamma \vdash B}{\Gamma \vdash A \wedge B}\,\wedge_R$$

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Sequent calculus – More inference rules

$$\frac{}{\Gamma \vdash \top}\,\top_R \qquad \frac{}{\Gamma, \bot \vdash C}\,\bot_L$$

$$\frac{\Gamma \vdash A}{\Gamma \vdash A \vee B}\,\vee_{\mathrm{R}_1} \quad \frac{\Gamma \vdash B}{\Gamma \vdash A \vee B}\,\vee_{\mathrm{R}_2} \qquad \frac{\Gamma, A \vee B, A \vdash C \qquad \Gamma, A \vee B, B \vdash C}{\Gamma, A \vee B \vdash C}\,\vee_{\mathrm{L}}$$

$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \to B}\,\to_{\mathrm{R}} \qquad \frac{\Gamma, A \to B \vdash A \qquad \Gamma, A \to B, B \vdash C}{\Gamma, A \to B \vdash C}\,\to_{\mathrm{L}}$$

Sequent calculus vs natural deduction

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- Natural deduction – Easier to reason with

Sequent calculus vs natural deduction

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- Sequent calculus – Easier to reason about

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$$\dfrac{A \vee \neg A \quad [A]^1 \quad \dfrac{\dfrac{\neg\neg A \quad [\neg A]^1}{\bot}\,\to\text{-elim}}{A}\,\bot\text{-elim}}{A}\,\vee\text{-elim}_1$$

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Decidabillity

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$\forall \Gamma, C. \quad \Gamma \vdash C \quad$ or $\quad \Gamma \nvdash C$

Decidabillity problem 1 – Termination

- Searching for proofs "bottom-up" can work well

$$\frac{}{A \wedge B \vdash B \wedge A}\ ?$$

Decidabillity problem 1 – Termination

- Searching for proofs "bottom-up" can work well

$$\dfrac{\dfrac{}{A \wedge B \vdash B}\,? \qquad \dfrac{}{A \wedge B \vdash A}\,?}{A \wedge B \vdash B \wedge A}\,\wedge_{\mathsf{R}}$$

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- Searching for proofs "bottom-up" can work well

$$\dfrac{\dfrac{\dfrac{}{A \wedge B, B \vdash B}\,?}{A \wedge B \vdash B}\,\wedge_{\mathsf{L}_2} \qquad \dfrac{}{A \wedge B \vdash A}\,?}{A \wedge B \vdash B \wedge A}\,\wedge_{\mathsf{R}}$$

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- But a computer can end up in an infinite loop...

$$\dfrac{}{A \wedge B \vdash B \wedge A}\,?$$

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- But a computer can end up in an infinite loop...

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Decidabillity problem 1 – Termination

- Naive solution: Detect infinite loops

[1] R. Dyckhoff, 1992, http://www.jstor.org/stable/2275431

Decidabillity problem 1 – Termination

- Naive solution: Detect infinite loops
- Solution: LJT / G4ip[1]

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Version of LJ without “infinite loops”

Solution: LJT / G4ip

Version of LJ without "infinite loops"

$$\frac{\Gamma, A \wedge B, B \vdash C}{\Gamma, A \wedge B \vdash C} \wedge_{L_2} \qquad \frac{\Gamma, A \wedge B, A \vdash C}{\Gamma, A \wedge B \vdash C} \wedge_{L_1}$$

Solution: LJT / G4ip

Version of LJ without "infinite loops"

$$\frac{\Gamma, A \wedge B, B \vdash C}{\Gamma, A \wedge B \vdash C} \wedge_{L_2} \qquad \frac{\Gamma, A \wedge B, A \vdash C}{\Gamma, A \wedge B \vdash C} \wedge_{L_1}$$

$$\downarrow$$

$$\frac{\Gamma, A, B \vdash C}{\Gamma, A \wedge B \vdash C} \wedge_{L}$$

Solution: LJT / G4ip

Version of LJ without "infinite loops"

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$$\downarrow$$

$$\frac{\Gamma, A, B \vdash C}{\Gamma, A \wedge B \vdash C} \wedge_L$$

$$\frac{\Gamma, A \vee B, A \vdash C \quad \Gamma, A \vee B, B \vdash C}{\Gamma, A \vee B \vdash C} \vee_L$$

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Version of LJ without "infinite loops"

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$$\downarrow$$

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Pathological case revisited

$$\dfrac{\dfrac{?}{A, B \vdash B \wedge A}\,?}{A \wedge B \vdash B \wedge A}\,\wedge_L$$

Decidabillity problem 2

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  Easy to devise an algorithm once termination is proven..
  difficult to prove in Agda
- Proof sketch **Case** $\Gamma = \Gamma', A \vee B$

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  1. I.H: Is there a derivation $\mathcal{D}$ of $\Gamma' A \vdash C$?

Decidabillity problem 2

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- Proof sketch **Case** $\Gamma = \Gamma', A \vee B$
  1. I.H: Is there a derivation $\mathcal{D}$ of $\Gamma' A \vdash C$?
  2. I.H: Is there a derivation $\mathcal{E}$ of $\Gamma', B \vdash C$?

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- Proof sketch **Case** $\Gamma = \Gamma', A \vee B$
  1. I.H: Is there a derivation $\mathcal{D}$ of $\Gamma' A \vdash C$?
  2. I.H: Is there a derivation $\mathcal{E}$ of $\Gamma', B \vdash C$?
  3. YES:

$$\dfrac{\begin{array}{c}\mathcal{D}\\ \Gamma', A \vdash C\end{array} \quad \begin{array}{c}\mathcal{E}\\ \Gamma', B \vdash C\end{array}}{\Gamma', A \vee B \vdash C} \vee_L$$

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- Goal: $\forall \Gamma, C. \quad \Gamma \vdash C \quad$ or $\quad \Gamma \nvdash C$
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- Proof sketch **Case** $\Gamma = \Gamma', A \vee B$
  1. I.H: Is there a derivation $\mathcal{D}$ of $\Gamma' A \vdash C$?
  2. I.H: Is there a derivation $\mathcal{E}$ of $\Gamma', B \vdash C$?
  3. YES:

$$\dfrac{\begin{array}{c}\mathcal{D}\\ \Gamma', A \vdash C\end{array} \quad \begin{array}{c}\mathcal{E}\\ \Gamma', B \vdash C\end{array}}{\Gamma', A \vee B \vdash C} \vee_L$$

  4. NO: If $\Gamma', A \nvdash C$ or $\Gamma', B \nvdash C$ then $\Gamma, A \vee B \nvdash C$

Decidabillity problem 2

4. NO: If $\Gamma', A \nvdash C$ or $\Gamma', B \nvdash C$ then $\Gamma, A \vee B \nvdash C$

Decidabillity problem 2

4. NO: If $\Gamma', A \nvdash C$ or $\Gamma', B \nvdash C$ then $\Gamma, A \vee B \nvdash C$

Proof sketch:

Assume $\Gamma', A \nvdash C$ (WOLOG), $\Gamma', A \vee B \vdash C$

Decidabillity problem 2

4. NO: If $\Gamma', A \nvdash C$ or $\Gamma', B \nvdash C$ then $\Gamma, A \vee B \nvdash C$

Proof sketch:

Assume $\Gamma', A \nvdash C$ (WOLOG), $\Gamma', A \vee B \vdash C$

Show: Contradiction!

Decidabillity problem 2

4. NO: If $\Gamma', A \nvdash C$ or $\Gamma', B \nvdash C$ then $\Gamma, A \vee B \nvdash C$

Proof sketch:

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Show: Contradiction!

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PROBLEM: Tooo many cases

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$$A_1, A_2, \cdots, A_n \mid A_{n+1}, \cdots, A_m \vdash C$$

"A_n is in front of the cursor"

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Also sequents with a cursor.

- $\Gamma \mid \Delta \vdash C$
- $\Gamma \mid \emptyset \vdash C$ "Cursor all the way to right"
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Here order matters! $A, B \mid \emptyset \vdash C \quad \not\equiv \quad B, A \mid \emptyset \vdash C$

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Move rules

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Right rules

$$\frac{\Gamma \vdash A}{\Gamma \vdash A \vee B} \vee_{R_1} \quad \rightarrow \quad \frac{\Gamma \mid \emptyset \vdash A}{\emptyset \mid \Gamma \vdash A \vee B} \vee_{R_1}$$

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Were we're at right now: $\forall \Gamma, C. \quad \Gamma \vdash C \quad$ or $\quad \Gamma \mid \emptyset \nvdash C$

Future work